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Task 2: Research on AI Virtual Fitting System

I researched how modern AI based virtual try-on systems work.
A virtual fitting system allows a person to see how a piece of clothing would look on them without physically wearing it. This is done entirely using computer vision and deep learning models. I studied the complete workflow of how these systems are built and what each stage does.

What is a Virtual Try-On System

A virtual try-on system takes two inputs , a photo of a person and a photo of a clothing item and produces an output image where the person appears to be wearing that clothing. It does not simply paste the clothing on top of the person. Instead it understands the body shape, adjusts the clothing to fit naturally, and blends it realistically.

Virtual Try-On Workflow

Stage 1 — Person Parsing

The body is divided into regions like face, torso, arms and legs.
Each pixel is labeled so the system knows what part of the body it is.

Stage 2 — Cloth Segmentation

The clothing item is separated from its background so only the garment remains as a clean image ready for the next stage.

Stage 3 — Warping

The flat garment image is stretched and reshaped to match the person's body pose and shape so it fits naturally.

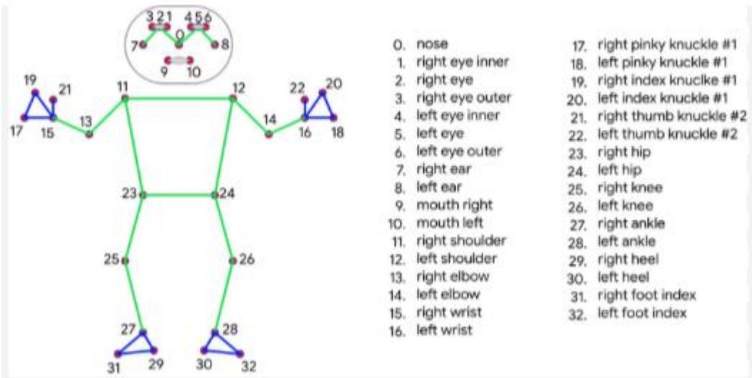
Stage 4 — Blending

The warped garment is merged onto the person's image using alpha blending to make the final output look realistic.

Pose Estimation

Pose estimation is the foundation of the entire virtual try-on system. Before any garment can be placed on a person the system needs to know exactly where the body is and how it is positioned. Pose estimation detects key body joints called landmarks such as shoulders, elbows, wrists, and hips. These landmark coordinates tell the system how to align and scale the garment correctly.

I used MediaPipe Pose for this task which detects 33 landmarks from a single image.



Segmentation

Segmentation divides the image into meaningful parts by classifying every pixel. For virtual try-on, the person is separated from the background using a body mask. This ensures the garment only appears on the body and not on the background. MediaPipe Selfie Segmentation is used for this which returns a confidence mask where 1 means person and 0 means background.

Garment Overlay

Garment overlay is placing the clothing item on the person's image after pose detection is done. The shoulder coordinates detected by MediaPipe are used to find the position and the shoulder width is used to scale the garment to the correct size. The garment is then placed at that position using alpha blending so it looks natural on the person.

Key Takeaway

Today I understood that a virtual try-on system is not just a simple image overlay. It involves multiple stages of computer vision working together — pose estimation to find the body, segmentation to separate body parts, warping to reshape the garment, and blending to make it look realistic. Each stage depends on the previous one which is why it is called a pipeline. Even a small error in pose estimation will affect all the stages that come after it. This research gave me a clear understanding of what needs to be built step by step for a complete virtual fitting system.